Class-II relative-phase source-budget obstruction

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1. Purpose and scope

Exact structure and a scoped no-go. The production Class-II matrix has two independent local phase-null directions and the actual A11 source is an exact output-shell commutator. These facts remove the scalar carrier that defeated A12's coefficient-blind estimate. They do not remove an internal $SU(2)$ relative-phase carrier. An explicit finite polynomial of degree 65536 gives the asymptotic exact-B source ratio $C_{\mathrm{rel}} = 0.91605\ldots > \frac{\gamma}{3} = 0.54$. Consequently the deterministic source-only sextic allocation fails for every Nelson exponent $p \geq 1$, including the A12 reference $p = 1.1$. The determinant resolvent tends to the identity on this carrier and does not repair it.

Tier boundary. This is a scoped T4 analytic/executed obstruction. It closes T-049 negatively. It does not rule out a joint source–potential log-Laplace cancellation, a different true-increment decomposition, probabilistic cancellation, changed parameters, a new filtration, or an interacting measure. It is not T5, T6, or T7.

The setting is the fixed $L = 16$ torus, three complex fields in the six-real convention, the hash-pinned A1 production symbol and Class-II coefficients, fixed $\rho_{\mathrm{floor}} = 10^{-12}$, and the A10–A12 strict dyadic rectangular cubes. Products are exact Fourier products. The regulator is the unit sharp cutoff member of the allowed common real-even family.

2. Exact Fierz reduction

Write $X = Z + W \in \mathbb{C}^2 \oplus \mathbb{C}$, with $Z = (z_1, z_2, 0)$ and $W = (0, 0, w)$. Put

$$s = |Z|^2, \qquad \rho = |X|^2, \qquad R = \rho + \rho_{\mathrm{floor}}.$$

For the embedded Pauli generators define

$$p_A = 2S_A X, \qquad q_A = \frac{X^T S_A X}{R}, \qquad v_A = p_A - 2q_A X.$$

The production coefficient matrix is

$$B(X) = \sum_A \{a p_A p_A^T + b(p_A v_A^T + v_A p_A^T) + c v_A v_A^T\}. \tag{2.1}$$

Set

$$d = a + 2b + c = 0.01059374999999735, \qquad \beta_{\mathrm{op}} = 4d = 0.04237499999999894.$$

Let E_Z be the projection onto the four real doublet coordinates and let $\mathcal{J}$ be the complex structure. Pauli completeness gives

$$P := \sum_A p_A p_A^T = 4\{sE_Z - (\mathcal{J}Z)(\mathcal{J}Z)^T\}, \tag{2.2}$$

$$\sum_A q_A p_A = 2\frac{s}{R}Z, \qquad \sum_A q_A^2 = \frac{s^2}{R^2}. \tag{2.3}$$

Substitution into (2.1) yields the exact fixed-floor matrix

$$\boxed{B(X) = dP - 4(b+c)\frac{s}{R}(Z \otimes X + X \otimes Z) + 4c\frac{s^2}{R^2}X \otimes X.} \tag{2.4}$$

In particular,

$$\boxed{B(X)\mathcal{J}Z = 0, \qquad B(X)\mathcal{J}W = 0.} \tag{2.5}$$

The two nulls are separate: one rotates the doublet by a common phase and the other rotates the singlet. They are stronger than the previously recorded single global-phase identity.

For a tangent $H = (\eta, \omega)$ let

$$ds = 2\Re(z^\dagger \eta), \qquad d\rho = ds + 2\Re(\bar{w}\omega).$$

The complex representative of $B(X)H$ is

$$\begin{aligned} B(X)H &= dP_J - 2(b+c)\frac{s}{R}\{Z\,d\rho + X\,ds\} + 2c\frac{s^2}{R^2}X\,d\rho, \\ P_J &= \left(4s\eta + 2(\eta^\dagger z - z^\dagger \eta)z, 0\right). \end{aligned} \tag{2.6}$$

Equivalently, for a spatial tangent $H = \partial_i X$,

$$B(X)H = \sum_A \{(aJ_A + bK_A)p_A + (bJ_A + cK_A)v_A\}, \tag{2.7}$$

where

$$K_A = R\,\partial_i q_A, \qquad v_A = R\nabla_X q_A. \tag{2.8}$$

Equations (2.4)–(2.8) are algebraic identities, not estimates.

They also sharpen the pointwise operator bound:

$$0 \le B(X) \le \beta_{\mathrm{op}} s \mathrm{I}, \qquad B(X)^2 \le \beta_{\mathrm{op}} s B(X). \tag{2.9}$$

Equality in the first bound is reached on an active doublet-horizontal tangent. Replacing s by the full ρ is valid but loses structure.

3. Local phase covariance and the exact shell object

Let $U_{\theta,\varphi}$ rotate the doublet by a spatially dependent phase $\theta(x)$ and the singlet by an independent phase $\varphi(x)$. Matrix covariance and (2.5) give

$$B(UX) = UB(X)U^T, \tag{3.1}$$

$$\partial_i(UX) = U\{\partial_i X + (\partial_i\theta)\mathcal{J}Z + (\partial_i\varphi)\mathcal{J}W\}, \tag{3.2}$$

and hence

$$\boxed{B(UX)\partial_i(UX) = UB(X)\partial_i X.} \tag{3.3}$$

Thus an arbitrary local common carrier in either phase sector contributes no derivative to the source.

Let $u_j = P_{\le j-1}\phi$ and let P_j be the disjoint next sharp cube shell. Since $P_j D u_j = 0$,

$$\boxed{P_j[B(u_j)Du_j] = [P_j, M_{B(u_j)}]Du_j.} \tag{3.4}$$

The actual source is

$$\widehat{\ell}_j(n) = R_\Lambda(n)A(n)^{-1/2}1_{\Sigma_j}(n)\{-ik_n \cdot \widehat{B(u_j)Du_j}(n)\}. \tag{3.5}$$

Dropping 1_{Σ_j} destroys the exact commutator and is forbidden by the A12 no-go.

The exact determinant resolvent also admits the identity

$$A_j = M_{B(u_j)}^{1/2} G_j, \quad T_j = A_j^* A_j, \quad K_j = A_j A_j^*,$$
$$r_j = M_{B(u_j)}^{1/2} Du_j, \quad \ell_j = A_j^* r_j,$$

$$\boxed{\langle \ell_j, (\mathrm{I} + pT_j)^{-1}\ell_j\rangle = \frac{1}{p}\{\|r_j\|^2 - \langle r_j, (\mathrm{I} + pK_j)^{-1} r_j\rangle\}.} \tag{3.6}$$

The second term is an exact negative rebate. Bounding (3.6) only by $\|r_j\|^2/p$ recreates the past-energy route already refuted by A11.

4. The active relative-phase carrier

The nulls (2.5) do not annihilate an internal $SU(2)$ relative phase. For

$$X = (e^{ik_1\cdot x} f, e^{ik_2\cdot x} g, 0)$$

with fixed amplitudes, the leading tangent has $ds = d\rho = 0$. Equation (2.6) therefore gives

$$B(X)D_i X = i\beta_{\mathrm{op}}(k_{1,i} - k_{2,i})\big(X_1|X_2|^2, -|X_1|^2 X_2, 0\big). \tag{4.1}$$

This is independent of ρ_{floor}. If each component contains only one Fourier mode, (4.1) remains in the past cube and the next-shell source is zero. A boundary modulation is required to move it into the output shell.

For a positive integer K, define the finite one-dimensional polynomial

$$f_K(z) = \sum_{n=1}^{K} n^{-5/6}(z^{-n} - z^n), \qquad u_K = P_- f_K = \sum_{n=1}^{K} n^{-5/6} z^{-n}. \tag{4.2}$$

On $\mathbb{T}^3$, put

$$F_K(x) = \prod_{a=1}^{3} f_K(e^{ix_a}), \qquad U_K(x) = \prod_{a=1}^{3} u_K(e^{ix_a}). \tag{4.3}$$

Choose a dyadic carrier $N > 3K$ and define

$$\Phi_N = \big(e^{iN(x_1+x_2+x_3)} F_K, e^{-iN(x_1+x_2+x_3)}\overline{F_K}, 0\big). \tag{4.4}$$

The previous sharp cube selects exactly

$$P_{\le N}\Phi_N = \big(e^{iN\sum_a x_a} U_K, e^{-iN\sum_a x_a}\overline{U_K}, 0\big). \tag{4.5}$$

Let

$$D_K = \|u_K\|_6^6, \qquad C_K = \|P_-(|u_K|^2 u_K)\|_2^2, \qquad B_K = \|f_K\|_6^6. \tag{4.6}$$

Tensorisation shows that the output cube keeps the complement of the nonpositive octant, with energy $D_K^3 - C_K^3$. The two complex components, the factor 8 in $\|\Phi_N\|_6^6$, the two opposite carrier momenta, and the q^{-4} covariance cancel to give

$$\boxed{\lim_{N\to\infty} \frac{\|G_j^* B(P_{\le N}\Phi_N) D(P_{\le N}\Phi_N)\|_2^2}{\|\Phi_N\|_6^6} = \beta_{\mathrm{op}}^2 \mathfrak{H}_{\mathrm{spin}}(K),} \tag{4.7}$$

$$\mathfrak{H}_{\mathrm{spin}}(K) = \frac{D_K^3 - C_K^3}{B_K^3}. \tag{4.8}$$

The physical $2\pi/L$ cancels in the high-frequency limit, and $Y = 1$ is retained from the production symbol.

5. Finite certificate and production comparison

Take

$$K = 65536. \tag{5.1}$$

The primary route computes the three finite coefficient convolutions with zero padding. The independent route evaluates the polynomial on an alias-free grid of 524288 points and projects the cubic by a separate FFT. They reproduce

$$\frac{D_K}{B_K} = 9.7254356828\ldots, \qquad \frac{C_K}{B_K} = 7.4272543964\ldots, \tag{5.2}$$

$$\mathfrak{H}_{\mathrm{spin}}(K) = 510.153710888\ldots, \tag{5.3}$$

and

$$\boxed{C_{\mathrm{rel}} = \beta_{\mathrm{op}}^2 \mathfrak{H}_{\mathrm{spin}}(K) = 0.9160527283\ldots.} \tag{5.4}$$

The recorded decision uses only the conservative inequalities

$$C_{\mathrm{rel}} > 0.9 > \frac{\gamma}{3} = 0.54. \tag{5.5}$$

For any $p \ge 1$,

$$\frac{\gamma}{3p} \le \frac{\gamma}{3} = 0.54, \tag{5.6}$$

so the necessary source-only absorption condition

$$C_{\mathrm{rel}} < \frac{\gamma}{3p} \tag{5.7}$$

fails for every admissible exponent. At the historical $p = 1.1$, the allowance is $0.4909090909\ldots$.

This obstruction is stronger than saying that a particular upper bound is too coarse. Equation (4.7) is a lower bound on the best exact-B source constant itself.

6. Why the resolvent does not repair the witness

For fixed K, the envelope in (4.2) is fixed while $N \to \infty$. From (2.9) and the production symbol,

$$\|T_j\| \le \beta_{\mathrm{op}} \|P_{\le N}\Phi_N\|_\infty^2 \sup_{q\in\Sigma_j} \frac{|q|^2}{\lambda_{\min} A(q)} = O_K(N^{-2}). \tag{6.1}$$

Consequently

$$\langle \ell_j, (\mathrm{I} + pT_j)^{-1} \ell_j \rangle \ge \frac{\|\ell_j\|^2}{1 + p\|T_j\|} = (1 - o(1))\|\ell_j\|^2. \tag{6.2}$$

Thus the same limit (4.7) holds with the exact determinant resolvent.

There is also an amplitude check. Scaling the entire field by δ leaves the source-to-sextic ratio invariant: both numerator and denominator scale as δ^6. Meanwhile $T_j = O(\delta^2)$. The relative carrier has $ds = d\rho = 0$, so its leading identity is floor-independent even as the amplitude is changed. The resolvent can therefore be made arbitrarily close to the identity without weakening the obstruction.

7. Consequence and new proof order

The gate

`A12-CLASSII-COEFFICIENT-AWARE-SHELL-LOCALISED-SOURCE-BOUND`

is closed negatively as a *production-budget* gate. A finite exact-B source bound may still exist, but no such bound can satisfy the current deterministic source-only sextic allocation.

The next gate is

`A13-CLASSII-JOINT-SOURCE-POTENTIAL-LOG-LAPLACE`.

It must change the proof architecture by retaining cancellations between the source and the local potential, changing the true-increment decomposition, or using a genuinely probabilistic cancellation. Merely improving a standalone source-square constant or restoring the resolvent is forbidden by (4.7)–(6.2).

This is recorded as

`NG-2026-07-21-A13-RELATIVE-PHASE-SOURCE-BUDGET`.

8. Devil's-advocate review

1. **Objection: the A12 common-phase null invalidates the witness. Dismissed.** The two components carry opposite phases. Equation (4.1) is an active $SU(2)$ horizontal tangent, not either null in (2.5).

2. **Objection: the nonlinear current remains in the past cube. Dismissed.** The Riesz boundary projection (4.5) and cubic product place exactly the complement of the nonpositive octant in the next shell. The carrier is chosen dyadic with $N > 3K$, so no terminal alias is used.

3. **Objection: the fixed floor changes the leading coefficient. Dismissed.** The carrier has $ds = d\rho = 0$. Every rational radial term in (2.6) vanishes, leaving the floor-independent dP_J term.

4. **Objection: a complex/six-real factor two or the torus scale was lost. Dismissed.** Both complex components, the factor eight in the sextic norm, $A(q)^{-1}$ for the six-real shell coordinates, and $2\pi/L$ are carried through (4.7). The two executable routes start from different coefficient-versus grid-space normalisations and agree.

5. **Objection: the finite calculation is a formal interval proof. Valid with mitigation.** It is not labelled as one. Two independent finite algorithms agree far inside the conservative inequality $C_{\mathrm{rel}} > 0.9$, leaving more than 0.36 above $\gamma/3$. The claim stays T4, and any cross-route or convergence failure is an explicit falsifier.

6. **Objection: the determinant resolvent restores the budget. Dismissed.** Equations (6.1)–(6.2) show $T_j \to 0$ on the fixed-envelope carrier. Amplitude scaling gives an independent limiting argument.

7. **Objection: the result proves the exact source is unbounded. Upheld.** No such statement is made. The result proves only that its best constant is larger than the production source-only budget.

8. **Objection: every possible Nelson or constructive route fails. Upheld.** The no-go is specific to deterministic standalone source absorption in the current true-increment allocation. Joint and redesigned routes remain open.

9. Reproduction and evidence matrix

Run from the repository root:

```
python codes/foundations/a13_classii_relative_phase_source_obstruction_verify.py
```

The command runs a primary coefficient-convolution audit, a non-importing alias-free grid audit, and integrated cross-checks. It writes three JSON artefacts under the A13 claim's `runs/` directory.

10. Result footer

```
Result ID:                A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Precise statement:        Exact B has two local phase nulls and a shell
                          commutator, but a K=65536 opposite-corner SU(2)
                          witness gives C_rel > 0.9 > gamma/3 = 0.54.
Scope:                    Fixed L=16 torus; A1 production coefficients; six-real
                          fields; fixed floor; unit sharp dyadic cubes.
Dependencies:             A1, A7, A9, A10, A11, and A12 claim packages.
Evidence grade:           ANALYTIC, EXACT, EXECUTED; independent reproduction.
Reproduction command:
  python codes/foundations/a13_classii_relative_phase_source_obstruction_verify.py
Expected output:          Primary 30/30; independent 18/18; integrated 77/77;
                          A13-...-OBSTRUCTION-INTEGRATED-PASS.
Falsification gate:       Any identity, source ratio, hash, PDF, assertion-count,
                          independent-route, or release-gate failure.
Tier before / after:      New claim / scoped T4; T-049 CLOSED-NEGATIVELY.
No-overclaim statement:   No source unboundedness, joint-route failure, Nelson
                          closure, interacting measure, T5, T6, or T7.
Next required action:     A13-CLASSII-JOINT-SOURCE-POTENTIAL-LOG-LAPLACE.
```

References

1. B. Hollenbeck and I. E. Verbitsky, "Best constants for the Riesz projection," *Journal of Functional Analysis* 175 (2000), 370–392, https://doi.org/10.1006/jfan.2000.3616.

2. TECT claim A12-CLASSII-SOURCE-SQUARE-REDUCTION, scalar-budget no-go and the exact-B successor gate.

3. TECT claim A11-CLASSII-TRUE-INCREMENT-DETERMINANT-REDUCTION, exact noncentral determinant and positive adapted source.

4. TECT claim A1-PRODUCTION-FUNCTIONAL-REALISATION, hash-pinned production coefficients and quadratic symbol.